Saminur Karim

Basic Scan (Scan of 1000 top ports)

Nmap Output | Ports / Hosts | Topology | Host Details | Scans

Port	Protocol	State	Service	Version
88	tcp	open	kerberos-sec	
445	tcp	open	microsoft-ds	
5000	tcp	open	upnp	
7000	tcp	open	afs3-fileserver	

Comprehensive Scan-4 More open TCP Ports (Scan of all 65,535 ports)

Nmap Output | Ports / Hosts | Topology | Host Details | Scans

Port	Protocol	State	Service	Version
88	tcp	open	kerberos-sec	
445	tcp	open	microsoft-ds	
5000	tcp	open	upnp	
7000	tcp	open	afs3-fileserver	
50798	tcp	open		
50842	tcp	open		
54610	tcp	open		
57621	tcp	open		

Service and Detection Scan- 2 Services unrecognized and Last 4 ports unknown services

Nmap Output | Ports / Hosts | Topology | Host Details | Scans

Port	Protocol	State	Service	Version
88	tcp	open	kerberos-sec	Heimdal Kerberos (server time: 2026-01-2
445	tcp	open	microsoft-ds	
5000	tcp	open	rtsp	
7000	tcp	open	rtsp	
50798	tcp	open		
50842	tcp	open		
54610	tcp	open		
57621	tcp	open		

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-31 17:43 -0500
Nmap scan report for mac.lan (192.168.86.37)
Host is up (0.000060s latency).
Not shown: 996 closed tcp ports (reset)
PORT     STATE SERVICE       VERSION
88/tcp   open  kerberos-sec  Heimdal Kerberos (server time: 2026-01-31
22:43:54Z)
445/tcp  open  microsoft-ds?
5000/tcp open  rtsp
7000/tcp open  rtsp
2 services unrecognized despite returning data. If you know the
service/version, please submit the following fingerprints at https://
nmap.org/cgi-bin/submit.cgi?new-service :
--------------NEXT SERVICE FINGERPRINT (SUBMIT
```

Service
kerberos-sec
microsoft-ds
rtsp
unknown

Hostname	Port	Protocol	State	Version
mac.lan (192.168.86.37)	54610	tcp	open	Unknown version
mac.lan (192.168.86.37)	56137	tcp	open	Unknown version
mac.lan (192.168.86.37)	56155	tcp	open	Unknown version
mac.lan (192.168.86.37)	57621	tcp	open	Unknown version

OS Detection-Inaccurate OS detection (Tahoe 26.2) and purpose type is correct

```
Host is up (0.00035s latency).
Not shown: 996 closed tcp ports (reset)
PORT     STATE SERVICE
88/tcp   open  kerberos-sec
445/tcp  open  microsoft-ds
5000/tcp open  upnp
7000/tcp open  afs3-fileserver
Device type: general purpose
Running: Apple macOS 12.X
OS CPE: cpe:/o:apple:mac_os_x:12
OS details: Apple macOS 12 (Monterey) (Darwin 21.1.0 - 21.6.0)
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://
nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.60 seconds
```

Aggressive Scan- Host Script results and includes all request information for the SYN packets. Includes content request, content length, request time, server name, etc.

```
PORT     STATE SERVICE       VERSION
88/tcp   open  kerberos-sec  Heimdal Kerberos (server time: 2026-01-31
22:49:15Z)
445/tcp  open  microsoft-ds?
5000/tcp open  rtsp
| fingerprint-strings:
|   FourOhFourRequest:
|     HTTP/1.1 403 Forbidden
|     Content-Length: 0
|     Server: AirTunes/925.5.1
|     X-Apple-ProcessingTime: 1
|     X-Apple-RequestReceivedTimestamp: 74967977
|   GetRequest:
|     HTTP/1.1 403 Forbidden
|     Content-Length: 0
|     Server: AirTunes/925.5.1
|     X-Apple-ProcessingTime: 1
|     X-Apple-RequestReceivedTimestamp: 74962931
|   HTTPOptions:
|     HTTP/1.1 403 Forbidden
|     Content-Length: 0
|     Server: AirTunes/925.5.1
|     X-Apple-ProcessingTime: 2
|     X-Apple-RequestReceivedTimestamp: 74967957
|   RTSPRequest:
|     RTSP/1.0 403 Forbidden
```

```
SF:den\r\nContent-Length:\x200\r\nServer:\x20AirTunes/925\.5\.1\r\nX-
Apple
SF:-ProcessingTime:\x201\r\nX-Apple-RequestReceivedTimestamp:
\x2076203982\
SF:r\n\r\n")%r(SIPOptions,A0,"RTSP/1\.0\x20403\x20Forbidden\r\nContent-
Len
SF:gth:\x200\r\nServer:\x20AirTunes/925\.5\.1\r\nCSeq:
\x2042\x20OPTIONS\r\
SF:nX-Apple-ProcessingTime:\x201\r\nX-Apple-RequestReceivedTimestamp:
\x207
SF:6204002\r\n\r\n");
Device type: general purpose
Running: Apple macOS 12.X
OS CPE: cpe:/o:apple:mac_os_x:12
OS details: Apple macOS 12 (Monterey) (Darwin 21.1.0 - 21.6.0)
Network Distance: 0 hops

Host script results:
| smb2-time:
|   date: 2026-01-31T23:33:15
|_  start_date: N/A
| smb2-security-mode:
|   3.0.2:
|_    Message signing enabled and required
|_clock-skew: -31s

OS and Service detection performed. Please report any incorrect results
at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 37.55 seconds
```

Report:

From the information in the scans, there were 4 open ports from the basic scan. The comprehensive scan located 8 open ports with unknown services. There were many protocols used for streaming audio/video (rtsp), server sharing, file sharing, and active directory control (Microsoft-DS), and secure network authentication and encryption (Kerberos). These are common protocols used by different end devices. The operating system detection was inaccurate as I am currently running a 26.2 Mac OS. The detection stated I have an outdated MacOS 12. There are many different threats and attacks that can be used against open port devices that are unprotected. Attacks like SYN flood and malicious port scanning attacks can discover open ports, vulnerabilities, and active services within a device. ICMP message abuse may also be utilized to DoS attack and ruin the network/device capability. Closed/Unused ports may also be susceptible to known service vulnerabilities or backdoors that allow unauthorized access. Brute-force attacks and password spraying attacks are also very common attacks

against ports. To see my exposed ports, I would look into my command prompt and close any unused ports. In order to secure my network, I would install anti-virus software to correct any backdoors installed. I would ensure that every device on my network is updated to the latest patch and to prevent any backdoors or vulnerablities. I would also check for any misconfigurations and default credentials for SSH. Changing my port services and protocols to secure ones with encryption is also a standard for network security. In terms of DoS attacks, I would secure a firewall and use packet filtering to allow only certain packets to enter through. Establishing proxy servers will also help to also segment my devices from malicious and suspicious IPs/web traffic. In case of a brute-force or password spraying attack, I would enforce strong password policies and IDS/IPS system to immedialetly alert or defend against such attack as a small measure. These multiple lines of defenses will secure my network and enforce the defense in depth strategy.

Brainstorming:
SYN flood attacks
Should have auditing software and watch/maintain logs
Malicious port scanning attacks
Should have network firewall active
ICMP message abuse
Should utilize packet filtering to manage against DoS attacks
Should also have anti-virus software to correct attacks
Close unused ports
Updated patches and software to protect against vulnerabilities
Regular port scanning with tools like Nmap
Packet filtering techniques